

Classwork Set - 1

1. Write a program to find the sum of n terms for the following series:

$$3/5 + 4/5^2 + 3/5^3 + 4/5^4 + 3/5^5 + 4/5^6 + \dots$$

2. Write a program to find the sum of n terms for the following series:

$$5 + 55 + 555 + 5555 + 55555 + \dots$$

3. Write a program to find the number of terms required for the sum of the following series to converge to a precision of 5 decimal points:

$$\frac{1^2}{2^3} + \frac{2^2}{3^3} + \frac{3^2}{4^3} + \frac{4^2}{5^3} + \dots$$

4. Write a program to find the sum of n terms of the series:

$$1 + 1 + 1 + 3 + 5 + 9 + 17 + 31 + 57 + 105 + \dots$$

5. Write a program to find the sum of n terms of the mod 17 Fibonacci series.

$$1 + 1 + 2 + 3 + 5 + 8 + 13 + 4 + 0 + 4 + 4 + 8 + \dots$$

6. Write a program to find the sum of n terms for the series:

$$1 + 2 + 4 + 7 + 11 + 16 + 22 + \dots$$

7. Write a program to find the sum of n terms for the series:

$$1 + 2 + 3 + 5 + 9 + 17 + 33 + 65 + \dots$$

Homework Set - 1

8. Write a program to find the number of terms required for the following series to form a cycle (i.e. the terms start repeating in the same pattern), and also find the sum of the terms in the cycle:

$$x \bmod 17 + x \bmod 23$$

9. Write a program to find the sum of n terms for the following series. If the series converges to a number, modify the program to generate terms to a 4 decimal place accuracy.

$$\sum_{n=2}^{\infty} \frac{1}{n^2 - 1}$$

10. Alice and Bob have invested Rupees 24270 and 41940 respectively in mutual funds A and B. A provides 2% compound interest whereas B provides 6% simple interest per month. Write a program to find whether the monthly interest of A can ever exceed B. Also find the number of months required for the total earned interest of both Alice and Bob to be same, with error tolerance of Rupees 500.